

Customer Segmentation and Value Analysis for a Retail Business

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This project applies data-driven analytics to segment retail customers based on their purchasing behavior, with the objective of supporting strategic decision-making in customer relationship management (CRM) and marketing. Using real transactional retail data from a publicly available dataset (UCI Machine Learning Repository), the analysis employs RFM (Recency, Frequency, Monetary) metrics and unsupervised machine learning techniques to identify actionable customer segments.

Data and Methodology

The dataset consists of over 500,000 transaction records from a retail business. After cleaning the data by removing missing customer identifiers, returned transactions, and pricing anomalies, RFM metrics were constructed to capture customer engagement and value. To ensure comparability across dimensions, monetary values were log-transformed and all RFM features were standardized. K-Means clustering was then applied, with the optimal number of clusters selected using a combination of the Elbow Method and Silhouette Analysis.

Key Findings

Four distinct customer segments were identified. A small but highly valuable group of High-Value Loyal Customers demonstrates very recent activity, high purchase frequency, and exceptionally high monetary value, representing a disproportionate share of total revenue. Growth Potential Customers show relatively frequent and recent purchases with moderate spending levels, indicating strong opportunities for upselling and loyalty development. At-Risk Customers are characterized by long periods of inactivity and low purchase frequency, signaling a high likelihood of churn. Finally, Low-Value or Occasional Customers exhibit infrequent purchases and limited spending, contributing marginally to overall revenue.

Managerial Implications

The segmentation results provide clear guidance for targeted retail strategies. High-Value Loyal Customers should be prioritized through personalized offers, loyalty programs, and exclusive benefits. Growth Potential Customers can be nurtured via targeted promotions, bundling, and cross-selling strategies to increase lifetime value. At-Risk Customers require win-back campaigns and reactivation incentives, while Low-Value Customers are best managed through cost-efficient, automated marketing approaches.